Conformal covariance of higher-dimensional exponential metrics

AI-assisted research note

Abstract

We formulate conformal covariance as a statement about a family of metric assignments on domains, rather than only about internal metrics of a whole-space field. The domain field is the zero-boundary fractional Gaussian field associated with the whole-space $\dot{H}^{d/2}$ energy. Given a strong whole-space metric assignment satisfying length-space, consistent locality, Weyl-scaling, and similarity-coordinate-change axioms, we construct its unique extension to locally LGF-type fields and prove conformal covariance in every dimension $d \geq 3$. The argument compares nearby conformal coordinate systems in probability, uses a shell estimate to obtain good annuli around every point, and improves a deterministic metric comparison constant to one. The whole-space assignment is an assumption; this paper does not establish its existence or identify an LFPP scaling limit.

1 Introduction

1.1 Fields and metric assignments on domains

A conformal coordinate-change theorem requires metrics for fields on different domains. We first specify this family. We then state the whole-space input from which the family is constructed. This separates the definition of a conformally covariant metric from the assertion that a given construction satisfies that definition, as in Gwynne–Miller [4].

Fix $d \geq 3$, $\xi > 0$, and $Q > \sqrt{2d}$. Set

$$\gamma = Q - \sqrt{Q^2 - 2d} \in (0, \sqrt{2d}), \qquad Q = \frac{d}{\gamma} + \frac{\gamma}{2}\,. \tag{1.1}$$

The Weyl exponent ξ and the coordinate-change exponent Q are parameters of the assumed whole-space assignment. We do not construct a function $\gamma \mapsto \xi$ in this paper. Parameters are fixed throughout, and we suppress the superscript γ on the metric.

For measurable assignments we use a fixed local Sobolev realization $H^{-m}_{\mathrm{loc}}(U)$, with an integer $m > d + 2$. This is a Polish space containing all fields used below almost surely. Statements about assignments are understood up to almost-sure equality for each fixed input law.

A whole-space log-correlated Gaussian field (LGF) H is a centered Gaussian distribution modulo additive constants with covariance

$$\mathbb{E}[(H, \rho)(H, \eta)] = \iint_{\mathbb{R}^d \times \mathbb{R}^d} \rho(x)\eta(y) \log \frac{1}{|x - y|}\, \mathrm{d}x\, \mathrm{d}y \tag{1.2}$$

for smooth compactly supported tests of integral zero. We choose a representative whenever a metric is evaluated. Changing this representative by c multiplies the metric by $e^{\xi c}$.

On a bounded domain U, the zero-boundary LGF $\mathring{h}^U$ is the Gaussian field with Cameron–Martin space

$$\mathcal{H}_0(U) = \overline{C_c^\infty(U)}^{\dot{H}^{d/2}(\mathbb{R}^d)}, \tag{1.3}$$

with the normalization specified in Section 2. In particular, this is not defined by taking a spectral fractional power of the Dirichlet Laplacian on U. It agrees with the usual zero-boundary GFF when $d = 2$.

We use a local field class so that the metric assignment is meaningful on every domain, without assuming that every domain supports a zero-boundary field with a fixed additive constant. A random distribution h on a domain U is *locally an LGF plus a continuous function* if, for every ball $B \Subset U$, it admits a coupling

$$h|_B = H|_B + f, \qquad f \in C(B), \tag{1.4}$$

where H is a whole-space LGF and f may depend on H. The coupling may depend on B. Restrictions of whole-space LGFs, and zero-boundary LGFs plus continuous functions, belong to this class; see Lemma 2.1.

For a continuous curve $P : [a, b] \to U$, define its length by

$$\operatorname{len}(P; D) = \sup_{a=t_0<\cdots<t_n=b} \sum_{j=1}^{n} D(P(t_{j-1}), P(t_j)). \tag{1.5}$$

For a continuous length metric D on a domain U and an open $W \subset U$, let $D(\cdot, \cdot; W)$ denote the infimum of these lengths over continuous paths in W between the specified endpoints. For $f \in C(U)$, define the Weyl multiplication

$$(e^{\xi f} \cdot D)(x, y) = \inf_{P:x\to y} \int_0^{\operatorname{len}(P;D)} e^{\xi f(P(t))}\, \mathrm{d}t, \tag{1.6}$$

where the paths lie in U and are parameterized by D-length. A continuous metric here means a finite metric inducing the Euclidean topology. All domains are open and connected; disconnected open sets can be treated component by component.

For a conformal diffeomorphism $\phi : U \to V$, write

$$J_\phi(x) = |\det \mathrm{D}\phi(x)|, \qquad h^\phi = h \circ \phi^{-1} + \frac{Q}{d} \log J_{\phi^{-1}}. \tag{1.7}$$

The composition is distributional: $(h \circ \phi^{-1}, \rho) = (h, (\rho \circ \phi)J_\phi)$. No independent recentering is performed after this change of coordinates.

Definition 1.1 (A domain family of LQG metrics)**.** A d-dimensional γ-LQG metric with Weyl exponent ξ is a family of measurable assignments $h \mapsto D_h$, one for each domain $U \subset \mathbb{R}^d$, with the following properties for locally LGF-type fields h on U.

I. *Length space.* Almost surely, D_h is a continuous length metric on U.

II. *Locality.* For each deterministic subdomain $W \subset U$, almost surely

$$D_{h|_W} = D_h(\cdot, \cdot; W)\,. \tag{1.8}$$

III. *Weyl scaling.* Almost surely, for every $f \in C(U)$,

$$D_{h+f} = e^{\xi f} \cdot D_h\,. \tag{1.9}$$

IV. *Conformal coordinate change.* For each deterministic conformal diffeomorphism $\phi : U \to V$, almost surely

$$D_h(x, y) = D_{h^\phi}(\phi(x), \phi(y)), \qquad x, y \in U\,. \tag{1.10}$$

The almost-sure event in (1.10) is simultaneous in the points, but may depend on ϕ and the fixed parameters.

The subscript h includes its domain. Thus D_h in this definition is a metric on the domain of h, not the restriction of an ambient distance function. Its construction below uses internal metrics, just as equation (1.10) of [4] does.

1.2 Whole-space input and main result

Hypothesis 1.2 (Strong whole-space assignment)**.** There is a Borel assignment $h \mapsto D_h$ on distributions on $\mathbb{R}^d$ satisfying the following conditions whenever h is a whole-space LGF plus a possibly dependent continuous function.

(i) Almost surely, D_h is a continuous length metric on $\mathbb{R}^d$.

(ii) For every deterministic domain W, there is a fixed Borel restriction rule $\mathcal{M}_W$ such that

$$D_h(\cdot, \cdot; W) = \mathcal{M}_W(h|_W) \quad \text{almost surely.} \tag{1.11}$$

The same rule is used for all the indicated whole-space inputs. For each such h and each fixed W, the identity is required on a common event after replacing h by $h + f$, simultaneously for $f \in C(\mathbb{R}^d)$.

(iii) Almost surely, simultaneously for $f \in C(\mathbb{R}^d)$, $D_{h+f} = e^{\xi f} \cdot D_h$.

(iv) For each deterministic similarity $S(x) = rRx + z$, with $r > 0$, $R \in O(d)$, and $z \in \mathbb{R}^d$, almost surely

$$D_h(Sx, Sy) = D_{h\circ S + Q\log r}(x, y), \qquad x, y \in \mathbb{R}^d\,. \tag{1.12}$$

The map-valued formulation (1.11) specifies the version of locality used in the domain construction. It is stronger as a formal statement than merely asserting, separately for each input law, the existence of some measurable factor. We assume these consistent, simultaneous versions; we do not derive them from that weaker wording. Rotations and reflections are included in (1.12). No uniqueness theorem for whole-space metrics is used.

To indicate the extension, couple a zero-boundary field $\mathring{h}^U$ with a whole-space LGF so that

$$H|_U = \mathring{h}^U + g_U, \qquad g_U \in C^\infty(U). \tag{1.13}$$

For a continuous f on U, the domain construction gives

$$D_{\mathring{h}^U + f} = e^{\xi(f - g_U)} \cdot D_H(\cdot, \cdot; U)\,. \tag{1.14}$$

The right side depends only on the distribution $\mathring{h}^U + f$, not on the coupling. Proposition 2.4 proves this assertion and constructs the family on all domains.

Theorem 1.3 (Conformal covariance)**.** *Under Hypothesis 1.2, the whole-space assignment has a unique extension to a domain family satisfying Definition 1.1. In particular, if h is locally an LGF plus a continuous function on U and $\phi : U \to V$ is a deterministic conformal diffeomorphism, then almost surely*

$$\boxed{D_h(x,y) = D_{h\circ\phi^{-1}+(Q/d)\log J_{\phi^{-1}}}\big(\phi(x),\phi(y)\big), \qquad x,y \in U.} \tag{1.15}$$

Uniqueness means almost-sure agreement for every fixed admissible input law, relative to the given normalization of the whole-space assignment.

This is an existence theorem for the domain family *conditional on* the strong whole-space assignment. It does not prove Hypothesis 1.2 for any higher-dimensional LFPP limit. In the planar theory the corresponding whole-plane input is a theorem, recalled as Theorem 1.2 of [4]; the conformal covariance theorem completes the verification of the domain-family definition. Here the same logical distinction is retained.

When $d = 2$ and ϕ is holomorphic, $J_\phi = |\phi'|^2$. Thus the correction in (1.15) becomes $Q \log |(\phi^{-1})'|$. Also, (1.3) becomes the Dirichlet-energy space for the zero-boundary GFF. These statements identify both the field and the coordinate-change formula with their planar counterparts. The proof below is for $d \geq 3$: it uses Liouville's theorem, which restricts conformal maps to Möbius maps. It is not a replacement for the planar proof for non-Möbius maps.

1.3 Outline of the proof

Section 2 defines the zero-boundary field, records the smooth-complement decomposition, and constructs the domain metrics. It proves Axioms I–III before using conformal covariance. In odd dimensions the precision operator is nonlocal, but the complementary field is still smooth in the domain. No ordinary boundary-value Markov property is required.

Section 3 compares a conformally transformed metric with the original metric on a small annulus. After normalization, the transformed fields have exactly the same Gaussian law and approach one another in probability. A measurable-function lemma then gives the required metric comparison, without assuming continuity of the metric as a function of an arbitrary field.

Section 4 applies the shell estimate of Contreras Hip–Zhuang [1, Lemma 2.7] to obtain good annuli around every point. Sphere connections give an initial deterministic comparison. A shortcut between two points on an inner sphere then forces a definite fraction of a geodesic's length into portions where the two metrics agree approximately. Any comparison constant greater than one can therefore be improved. The reverse inequality follows by applying the one-sided result to the inverse coordinate change and then using Weyl scaling.

We write $B_r(z)$ for a Euclidean ball and $A_{a,b}(z) = B_b(z) \setminus \overline{B_a(z)}$. The notation $K \Subset U$ means that K is compact and contained in U. Internal distances are infima of lengths of continuous paths; nonrectifiable paths have infinite length.

2 Zero-boundary fields and the domain construction

2.1 The zero-boundary LGF

Use the unitary Fourier transform with phase $e^{-ix\cdot\zeta}$, and put

$$\kappa_d = \frac{2^{1-d}\pi^{-d/2}}{\Gamma(d/2)}, \qquad \mathcal{E}(f,g) = \kappa_d \int_{\mathbb{R}^d} |\zeta|^d \widehat{f}(\zeta)\overline{\widehat{g}(\zeta)}\,\mathrm{d}\zeta. \tag{2.1}$$

This is the normalization for which the whole-space covariance is (1.2); see [1, Section 2.1]. In particular, $\mathcal{E}(f,g) = (2\pi)^{-1}\int \nabla f \cdot \nabla g$ when $d = 2$.

For a bounded domain U, complete $C_c^\infty(U)$ under $\mathcal{E}$ and denote the resulting real Hilbert space by $\mathcal{H}_0(U)$. Elements are extended by zero when their whole-space energy is evaluated. The zero-boundary LGF $\mathring{h}^U$ is the centered Gaussian distribution with this Cameron–Martin space. More explicitly, if $(e_j)_{j\ge1}$ is an orthonormal basis of $\mathcal{H}_0(U)$ and $(a_j)_{j\ge1}$ are independent standard normal variables, then

$$\mathring{h}^U = \sum_{j\ge1} a_j e_j \tag{2.2}$$

in the sense of distributions. Its law is independent of the basis. Equivalently, if $u_\rho \in \mathcal{H}_0(U)$ satisfies $\mathcal{E}(u_\rho, v) = \int_U \rho v$ for every $v \in \mathcal{H}_0(U)$, then

$$\mathbb{E}[(\mathring{h}^U,\rho)(\mathring{h}^U,\eta)] = \mathcal{E}(u_\rho,u_\eta). \tag{2.3}$$

The construction, including the continuity of these test-function pairings, is given in [5, Sections 4–5].

Thus the relevant precision operator is a constant multiple of $(-\Delta)^{d/2}$, with boundary conditions specified by the completion, not merely by the name of the operator. On a smooth domain in dimension four these are the clamped biharmonic conditions. In odd dimensions the energy is nonlocal. Taking a spectral power of the Dirichlet Laplacian on U generally gives a different field. On an unbounded domain the zero-boundary construction requires the corresponding test-function continuity condition (an allowable domain in the terminology of [5, 1]); our local definition (1.4) does not require this condition on U itself.

Lemma 2.1 (Smooth complementary field)**.** *Let $U \subset \mathbb{R}^d$ be bounded. A whole-space LGF modulo constants admits an orthogonal Gaussian decomposition whose restriction to U has the form*

$$H|_U = \mathring{h}^U + g_U, \qquad g_U \in C^\infty(U) \quad \text{almost surely.} \tag{2.4}$$

The complementary full-space distribution is $d/2$-harmonic in U. The two Gaussian components are independent before choosing an additive constant, and they can be represented independently by normalizing the complementary component using an average supported outside $\overline{U}$.

If $W \subset U$ is a bounded subdomain, the analogous orthogonal decomposition gives

$$\mathring{h}^U|_W = \mathring{h}^W + g_{W,U}, \qquad g_{W,U} \in C^\infty(W), \tag{2.5}$$

with independent summands. Consequently, a zero-boundary LGF plus a continuous function is locally an LGF plus a continuous function.

Proof. The closed subspace $\mathcal{H}_0(U)$ of the whole-space Cameron–Martin space defines the orthogonal projection of the Gaussian field. The projected field has covariance (2.3). Its orthogonal complement annihilates $(-\Delta)^{d/2}\varphi$ for $\varphi \in C_c^\infty(U)$. The fractional Gaussian field decomposition and interior smoothness theorem are [5, Section 5 and Theorem 8.3]. Applying the same projection to $\mathcal{H}_0(W) \subset \mathcal{H}_0(U)$ gives (2.5). For the last assertion, rearrange (2.4) on any relatively compact ball. □

The equation $(-\Delta)^{d/2} g_U = 0$ in odd dimensions refers to the full-space complementary distribution. It is not an equation determined by the values of g_U in U alone. Also, subtracting a normalization average supported inside U from H can destroy the independence in (2.4); it only changes g_U by a random constant and does not affect its smoothness. The domain construction uses smoothness, not independence after an arbitrary normalization.

Lemma 2.2 (Conformal invariance of the zero-boundary field)**.** *Let $U, V \subset \mathbb{R}^d$ be bounded domains and let $\phi : U \to V$ be conformal. Then*

$$\mathring{h}^U \circ \phi^{-1} \overset{\text{law}}{=} \mathring{h}^V. \tag{2.6}$$

Proof. By Liouville's theorem, ϕ is the restriction of a Möbius transformation. The fractional-Laplacian transformation formula in [2, Lemma 3], at the critical order $s = d/2$, reads

$$(-\Delta)^{d/2}(v \circ \phi) = J_\phi\big((-\Delta)^{d/2} v\big) \circ \phi. \tag{2.7}$$

For $v \in C_c^\infty(V)$, its support avoids the image of infinity under the global Möbius map: that point does not belong to $\phi(U)$ when U is bounded. Thus $v \circ \phi$, extended by zero, has compact support in U and is smooth away from, and identically zero near, the pole. These are the functions used in the energy calculation. Pairing (2.7) with $u \circ \phi$ and changing variables gives

$$\mathcal{E}(u \circ \phi, v \circ \phi) = \mathcal{E}(u, v), \qquad u, v \in C_c^\infty(V). \tag{2.8}$$

Composition therefore extends to an isometry from $\mathcal{H}_0(V)$ onto $\mathcal{H}_0(U)$. Transforming the orthonormal series (2.2) proves (2.6). □

This lemma concerns the Gaussian input, not the metric. The metric coordinate-change formula still has to be proved. In dimension two the same conclusion follows from invariance of the Dirichlet energy under arbitrary planar conformal maps.

2.2 Local length structures

We record the deterministic facts used to pass between local and global metrics. For any continuous length metric D on a domain U, the internal metric on a subdomain W is finite and induces the Euclidean topology. Indeed, near a point $x \in W$, choose a Euclidean ball with closure in W. Its boundary is a compact set at positive D-distance from x. Every sufficiently short almost-minimizing path stays in that ball. Internal and ambient distances consequently agree near x. The set of points reachable by finite-length paths in W is then both open and relatively closed, and hence is all of W.

For a curve contained in W, its length computed using D is the same as its length computed using $D(\cdot, \cdot; W)$. This follows by subdividing the curve into the neighborhoods just described. Thus internalization is associative, and it commutes with Weyl multiplication:

$$(e^{\xi f} \cdot D)(\cdot, \cdot; W) = e^{\xi f|_W} \cdot D(\cdot, \cdot; W). \tag{2.9}$$

The weighted metric in (1.6) is again a continuous length metric. Its induced length on a rectifiable curve is its weighted D-length. These assertions follow locally from the upper and lower bounds for $e^{\xi f}$ on compact sets, followed by subdivision.

Lemma 2.3 (Patching local metrics)**.** *Suppose a countable collection of relatively compact balls covers U. Assign to each ball B a continuous length metric d_B. Assume that these metrics have the same internal metrics on every smaller ball contained in an overlap. Then there is a unique continuous length metric D on U with these local length structures. It satisfies $D(\cdot, \cdot; B) = d_B$ on each ball of the collection.*

If the local metrics are Borel functions of a distribution restricted to their balls, the patched metric is a measurable function of that distribution.

Proof. For a continuous path P, subdivide its parameter interval so that each piece lies in one covering ball. Sum the lengths of the pieces in the corresponding metrics. This sum does not depend on the subdivision or the balls: a common refinement reduces the assertion to a smaller ball in an overlap, where internalization preserves length. Define D to be the infimum of these lengths over paths between the two endpoints.

Finite chains of overlapping balls, joined using finite-length paths in each ball, show that D is finite. To check nondegeneracy and topology, fix $x \in B$ and concentric closed balls $C_1 \subset \mathrm{int}(C_2) \subset C_2 \subset B$ centered at x. The number $a = d_B(C_1, \partial C_2)$ is positive. A path starting in C_1 and leaving C_2 incurs prescribed length at least a before its first exit. Choose a neighborhood $N \subset C_1$ of x with $\sup_{u,v\in N} d_B(u,v) < a/2$. For $u, v \in N$, almost-minimizing d_B-paths of length less than a remain in C_2, whereas any competing path leaving C_2 costs at least a. A path remaining there has its prescribed d_B-length. Hence $D(u,v) = d_B(u,v)$ for all $u, v \in N$. This proves that D is a metric inducing the Euclidean topology. It also shows that its path lengths are the prescribed lengths. By construction it is a length metric, and internalizing it on B recovers d_B. The same argument gives uniqueness.

For measurability, fix a countable dense set in U. Distances between its points can be written as infima of sums of local distances along finite chains whose intermediate vertices belong to this set and whose successive vertices lie in a covering ball. The chain formula agrees with the path definition: one direction uses near-minimizing local paths; the other subdivides a path and moves its finitely many subdivision points within the overlaps. All terms in the chain formula are measurable. The evaluation map from $C(U \times U)$ to the sequence of values on this countable set is Borel and injective. Its image is Borel and its inverse on that image is Borel. Applying this inverse to the measurable chain distances gives the claimed map; define an arbitrary fixed continuous function when the sequence is outside the image. On every admissible input law the construction lies in the image almost surely. □

2.3 Extension of the whole-space assignment

Proposition 2.4 (Construction of domain metrics)**.** *Under Hypothesis 1.2, there is a unique extension of the whole-space assignment to locally LGF-type fields satisfying Axioms I–III of Definition 1.1. For a bounded domain U and the decomposition (2.4), this extension satisfies (1.14).*

Proof. Fix a field h on U. Choose a countable collection of balls B with slightly larger concentric balls $O \Subset U$ that covers U, and include a countable basis of smaller balls in overlaps. On O, couple $h = H + f$ as in (1.4). Choose a smooth cutoff χ supported in O and equal to one near $\overline{B}$. Extend χf by zero to $\mathbb{R}^d$. The whole-space field $H + \chi f$ is an admissible input to Hypothesis 1.2. Therefore

$$d_B := \mathcal{M}_B(h|_B) = D_{H+\chi f}(\cdot,\cdot;B) = e^{\xi f|_B} \cdot D_H(\cdot,\cdot;B) \tag{2.10}$$

almost surely. The fixed restriction rule is what makes the left side independent of the auxiliary coupling and cutoff.

For a smaller ball $A \subset B$, associativity of internalization and (1.11), applied to the same whole-space extension, give

$$d_B(\cdot,\cdot;A) = \mathcal{M}_A(h|_A). \tag{2.11}$$

All required identities for the countable collection hold on a common event. Lemma 2.3 now constructs a measurable continuous length metric D_h on U.

To establish locality for a fixed subdomain W, use countable ball collections for U and W together. The two constructions have the same lengths on every path in W, by (2.11); their internal distances

are therefore equal. If h is globally defined, the same argument identifies D_h with the original whole-space metric.

Weyl scaling follows first on each ball from the whole-space Weyl identity applied to a cutoff extension of the added continuous function. The whole-space identity is simultaneous in that function. Taking the countable collection of balls and using (2.9) gives the corresponding simultaneous identity on U. This also handles continuous perturbations dependent on the base field.

For (1.14), use $h = \mathring{h}^U + f = H|_U + (f - g_U)$ and (2.10). The function $f - g_U$ need not extend continuously across ∂U: its cutoff on each relatively compact ball is sufficient. Equality of the resulting local path lengths gives the formula on U.

Finally, any other family agreeing with the whole-space assignment and satisfying locality and Weyl scaling must satisfy (2.10). Lemma 2.3 then identifies the two families. Uniqueness, like the axioms, is an almost-sure assertion for each fixed admissible input law. □

No boundary-value Markov property was used in this construction. The Gaussian decomposition supplies the relation between zero-boundary fields and the local field class; fixed local metric rules and Weyl scaling supply the metric extension.

3 Comparison near a similarity

For a conformal map $\phi : U \to V$, pull the transformed domain metric back to U:

$$D_h^\phi(x, y) = D_{h^\phi}(\phi(x), \phi(y)), \qquad x, y \in U. \tag{3.1}$$

It is a continuous length metric. Its internal metric on $W \subset U$ is the pullback of $D_{h^\phi}(\cdot, \cdot; \phi(W))$. All comparisons in this section precede the proof of conformal covariance.

3.1 The field under a nearby coordinate change

Lemma 3.1 (Local Möbius invariance)**.** *Let ψ be a Möbius transformation smooth on an open set B. The restriction $H \circ \psi|_B$, modulo constants, has the law of the restriction of a whole-space LGF. Suppose ψ_n converges in the Möbius group to a similarity S. On each bounded smooth domain, subtracting a fixed smooth average gives fields with identical laws, and these fields converge in probability to the corresponding normalized restriction of $H \circ S$.*

Proof. For similarities and inversion, and hence for their compositions,

$$|\psi(x) - \psi(y)| = |x - y|\big(J_\psi(x) J_\psi(y)\big)^{1/(2d)}. \tag{3.2}$$

The transformed logarithmic kernel thus differs from the original kernel by a sum of a function of x and a function of y. Such terms vanish against two mean-zero tests. The centered Gaussian laws modulo constants agree; this is the invariance recorded in [3, Proposition 1.2].

On a fixed bounded set, the poles of ψ_n eventually lie outside a neighborhood of its closure, and the maps converge smoothly there. Distributional pullbacks consequently converge. More concretely, use the realization in a sufficiently negative local Sobolev space, restrict from a slightly larger bounded set, and use continuity of pullback by smoothly converging diffeomorphisms. Normalization by a compactly supported smooth average is continuous. This gives convergence in probability in a Polish space of distributions, for example a sufficiently negative local Sobolev space. The first part gives equality of all the normalized laws. □

The lemma also proves that the field class (1.4) is preserved by (1.7). A normalized pullback restriction can be coupled to a normalized whole-space LGF restriction. Restoring the average adds only a random constant, and the Jacobian correction is smooth. The assertion is local; no extension across a pole is required.

Lemma 3.2 (Measurable functions of nearby equal-law inputs)**.** *Let X_n, X take values in a Polish space, have the same law, and satisfy $X_n \to X$ in probability. If F is a Borel map into another Polish space, then $F(X_n) \to F(X)$ in probability.*

Proof. For any $\eta > 0$, Lusin's theorem and tightness give a compact set of probability at least $1 - \eta$ on which F is uniformly continuous. The probability that either input lies outside this set is at most 2η. On the set, convergence of the inputs implies convergence of the outputs in probability. Let $\eta \downarrow 0$. □

We will also need to control the domain in which paths are allowed to run.

Lemma 3.3 (Internal upper semicontinuity)**.** *Let d_n, d be continuous length metrics on a common open set B, with $d_n \to d$ locally uniformly on $B \times B$. Let W be a bounded smooth domain and suppose ψ_n are diffeomorphisms on a neighborhood of $\overline{W}$, converging there smoothly to a similarity S, with $S\overline{W} \Subset B$. For $K \Subset W$,*

$$\limsup_{n\to\infty} \sup_{u,v\in K} \big[d_n(\psi_n u, \psi_n v; \psi_n W) - d(Su, Sv; SW)\big] \le 0. \tag{3.3}$$

Proof. First let $u_n \to u$ and $v_n \to v$ in K. Choose a d-rectifiable path in SW from Su to Sv, with length within η of the internal distance. Its image is compact in SW. Partition it into finitely many pieces whose lengths are smaller than a fixed fraction of the d-distance of the image from $\partial(SW)$.

Move each subdivision point by $\psi_n \circ S^{-1}$. The d_n-distance between each consecutive pair converges to its d-distance. Moreover, the d_n-distance from each moved point to $\partial(\psi_n W)$ stays bounded below by a larger positive number: the boundaries converge in a fixed compact subset of B, and the metrics converge there uniformly. An almost-minimizing d_n-path between the consecutive points therefore cannot exit $\psi_n W$. Concatenating these paths gives the desired upper bound. The moved endpoints can be joined to $\psi_n u_n$ and $\psi_n v_n$ inside $\psi_n W$ at cost tending to zero by the same argument.

Let $n \to \infty$ and then $\eta \downarrow 0$. The internal metric of d is continuous on SW. Compactness of K upgrades the sequential assertion to (3.3). □

3.2 A uniform small-scale estimate

Proposition 3.4 (Near-similarity comparison)**.** *Let ϕ_n be Möbius maps converging to a similarity S in the Möbius group. Let W be a bounded smooth domain and $K \Subset W$. For every $\delta > 0$,*

$$\mathbb{P}\left[\sup_{u,v\in K} \big(D_H^{\phi_n}(u, v; W) - D_H(u, v; W)\big) > \delta\right] \longrightarrow 0. \tag{3.4}$$

The restrictions in this formula are taken only where the maps are smooth, which holds for all sufficiently large n.

Proof. Choose a bounded smooth domain B containing $S\overline{W}$ and $\phi_n\overline{W}$ for all sufficiently large n. The inverse maps are smooth on a neighborhood of $\overline{B}$ for all such n. Set

$$g_n = H \circ \phi_n^{-1}, \quad g = H \circ S^{-1}, \quad m_n = (g_n, \rho), \quad m = (g, \rho), \tag{3.5}$$

where $\rho \in C_c^\infty(B)$ has integral one. By Lemma 3.1, the fields $\widehat{g}_n = g_n - m_n$ and $\widehat{g} = g - m$ have the same law and converge in probability on B. The domain assignment is Borel, with values in the Polish space $C(B \times B)$ equipped with local uniform convergence. Lemma 3.2 gives

$$D_{\widehat{g}_n} \longrightarrow D_{\widehat{g}} \quad \text{in probability, locally uniformly on } B \times B. \tag{3.6}$$

Also $m_n \to m$ in probability. If S has scale factor r, then

$$q_n := \frac{Q}{d} \log J_{\phi_n^{-1}} \longrightarrow -Q \log r \tag{3.7}$$

uniformly on B. Put $c = m - Q \log r$ and $a_n = \sup_B |m_n + q_n - c|$. Then $a_n \to 0$ in probability, and Weyl scaling gives

$$e^{-\xi a_n} e^{\xi c} D_{\widehat{g}_n} \le D_{g_n + q_n} \le e^{\xi a_n} e^{\xi c} D_{\widehat{g}_n}. \tag{3.8}$$

Together with (3.6), these bounds give convergence to $e^{\xi c} D_{\widehat{g}} = D_{g - Q \log r}$ locally uniformly in probability. Here the limiting Weyl multiplier is constant; no continuity assertion for arbitrary perturbations of a field is being assumed.

Along an almost surely convergent subsequence, apply Lemma 3.3 to these metrics and the maps ϕ_n. The resulting limiting internal distance is

$$D_{H \circ S^{-1} - Q \log r}(Su, Sv; SW) = D_H(u, v; W) \tag{3.9}$$

by the similarity axiom and locality. Every subsequence has a further subsequence on which these steps hold almost surely. This proves (3.4). □

Write $H_r(z)$ for the spherical average and put

$$s_r(z) = r^{\xi Q} e^{\xi H_r(z)}, \qquad H^{z,r} = H(z + r\,\cdot) - H_r(z). \tag{3.10}$$

The normalized field $H^{z,r}$ has a law independent of z, r. The similarity and constant-Weyl axioms give

$$s_r(z)^{-1} D_H(z + r\,\cdot, z + r\,\cdot) = D_{H^{z,r}}. \tag{3.11}$$

Spherical averages and their scaling law for this normalization are discussed in [1, Section 2.1].

Corollary 3.5 (Uniform comparison on small annuli)**.** *Fix $\phi : U \to V$ conformal, $K_0 \Subset U$, a bounded annulus W centered at zero, and $K \Subset W$. For every $\delta > 0$,*

$$\sup_{z \in K_0} \mathbb{P}\left[\sup_{u,v \in K} \left(D_H^{\phi}(z + ru, z + rv; z + rW) - D_H(z + ru, z + rv; z + rW) \right) > \delta s_r(z) \right] \longrightarrow 0 \tag{3.12}$$

as $r \downarrow 0$. Only radii for which $z + r\overline{W} \subset U$ are used.

Proof. Set

$$\phi_{z,r}(u) = \frac{\phi(z + ru) - \phi(z)}{r}. \tag{3.13}$$

For any sequences $z_n \in K_0$ and $r_n \downarrow 0$, a subsequence has $z_n \to z$ and $\phi_{z_n, r_n} \to \mathrm{D}\phi(z)$ smoothly on every fixed bounded set. The limit is a similarity. The two metrics in (3.12), divided by $s_r(z)$, are respectively $D_{H^{z,r}}^{\phi_{z,r}}$ and $D_{H^{z,r}}$, by the two similarity coordinate changes and the Jacobian chain rule. Since $H^{z,r}$ has a fixed law, Proposition 3.4 applies to this subsequence. A contradiction argument proves uniformity in z. □

4 Comparison by annular connections

Fix a conformal map $\phi : U \to V$ and a whole-space LGF H. In this section write

$$D = D_H \quad \text{on } \mathbb{R}^d, \qquad D^U = D_H(\cdot,\cdot;U), \qquad E = D_H^\phi \quad \text{on } U. \tag{4.1}$$

The objective is to prove $E \le D^U$. We use the open shells

$$A(z,r) = A_{r/4,4r}(z). \tag{4.2}$$

The spheres of radii $r/2$, αr, and r, for $1/2 < \alpha < 1$, all lie strictly inside this shell. If $1/4 < a < b < 4$ and $\overline{A(z,r)} \subset U$, then

$$\begin{aligned} D(\partial B_{ar}(z), \partial B_{br}(z)) &= D(\partial B_{ar}(z), \partial B_{br}(z); U) \\ &= D(\partial B_{ar}(z), \partial B_{br}(z); A(z,r)). \end{aligned} \tag{4.3}$$

Indeed, in any path from the inner to the outer sphere, take the last visit to the inner sphere before the first visit to the outer sphere. The intervening subpath lies in the closed radial region between the spheres, hence in $A(z,r)$, and is no longer than the original path. Taking infima proves (4.3). The same subpath argument applies to any length metric already defined on U.

4.1 Good shells at every point

We use the following consequence of [1, Lemma 2.7]. For any $a > 0$, there is $p < 1$ such that, if $r_{k+1} = r_k/16$ and each event F_k is measurable with respect to

$$(H - H_{r_k}(z))|_{A(z,r_k)}, \tag{4.4}$$

with $\mathbb{P}[F_k] \ge p$, then

$$\mathbb{P}[\text{none of } F_1, \dots, F_N \text{ occurs}] \le e^{-aN}. \tag{4.5}$$

The spherical average in (4.4) is measurable from the open shell: it is obtained as a limit of smooth averaging tests supported in successively smaller neighborhoods of the sphere, all inside that shell. We do not take the trace of an arbitrary distribution on a sphere.

The estimate follows from the cited lemma with radius ratio $1/16$, annulus parameter 4, and required proportion of successes $1/2$. The parameter p can be chosen arbitrarily close to one as needed. We only invoke the stated bound, not independence of different shells.

Lemma 4.1 (Covering by successful balls)**.** *There is $p_* = p_*(d) < 1$ with the following property. Suppose events $F_{z,r}$ are measurable with respect to (4.4) and, for each compact set of centers in U, have probability at least p_* for all sufficiently small r, uniformly over those centers. Almost surely, for each fixed $K \Subset U$ and all sufficiently small dyadic ε, every point of K belongs to $B_{r/2}(z)$ for some successful $F_{z,r}$, where*

$$\varepsilon^2 \le r \le \varepsilon, \qquad r \in \{16^{-k} : k \in \mathbb{Z}\}, \tag{4.6}$$

and z belongs to a deterministic lattice of mesh ε^4 in a fixed compact neighborhood of K contained in U.

Proof. Choose $a > (4d+3)\log 16$ in (4.5) and use the associated threshold p_*. The number of radii in (4.6) is at least $(\log \varepsilon^{-1})/\log 16 - O(1)$. For each lattice center, the probability of no successful radius is therefore $O(\varepsilon^{4d+3})$. There are $O(\varepsilon^{-4d})$ centers. A union bound gives failure probability $O(\varepsilon^3)$, which is summable over dyadic ε. Borel–Cantelli applies. The distance to the nearest lattice point is $O(\varepsilon^4) = o(\varepsilon^2)$, so the successful half-balls cover K. The same conclusion holds simultaneously on a countable compact exhaustion of U. □

The following domain comparison is proved directly, rather than inferred from a whole-space comparison theorem.

Lemma 4.2 (Sphere-chain comparison)**.** *Let D_1, D_2 be continuous length metrics on U. Suppose that, for some deterministic $C > 0$, the events*

$$\sup_{u,v\in\partial B_r(z)} D_2(u, v; A(z, r)) \le CD_1(\partial B_{r/2}(z), \partial B_r(z); U) \tag{4.7}$$

are measurable with respect to (4.4) and satisfy the probability hypothesis of Lemma 4.1. Then almost surely $D_2 \le CD_1$ on $U \times U$.

Proof. Work on the event of Lemma 4.1 for a compact exhaustion. Let P be any D_1-rectifiable path, parameterized by length on $[0, T]$, with endpoints x, y and image in a compact subset of U. For a successful covering at scale ε, start at $t_0 = 0$. Choose a successful ball whose half-ball contains $P(t_0)$, and follow P until its first exit from the outer ball. Repeat from the exit point, stopping at T if there is no further exit. Each completed step has duration at least the D_1-distance between the two spheres of its chosen ball. There are finitely many available balls, and their crossing distances have a positive minimum. Hence there are finitely many steps.

Consider the outer balls of the completed steps. Consecutive balls intersect, since the exit point from one lies in the half-ball of the next. Their union is connected. Remove balls contained in another one. The remaining union is unchanged, and the intersection graph of the remaining balls is connected. Two intersecting open Euclidean balls neither of which contains the other have intersecting boundary spheres. It follows that the union of these boundary spheres is connected. This is the elementary sphere construction used in [1, proof of Proposition 3.13].

Choose a simple chain of these spheres from a sphere at Euclidean distance at most 2ε from x to one at distance at most 4ε from y. Such spheres exist: a maximal ball contains the first completed ball, and a maximal ball contains the last completed ball; the remaining terminal path lies in a ball of radius at most ε. If there is no completed step, then $|x - y| \le 2\varepsilon$, a case that disappears as $\varepsilon \downarrow 0$ for fixed distinct endpoints.

For each sphere in the simple chain, connect the entrance point to the exit point using an almost-minimizing D_2-path in its shell. By (4.7) this costs at most C times the crossing distance for that ball, plus an arbitrarily small error. Assign that ball to one of its completed steps. Distinct balls are assigned distinct steps, so the sum of the crossing distances is at most T. All shells lie in a fixed compact neighborhood of the image of P for small ε. By continuity of D_2, the costs of joining the two ends of the sphere chain to x, y tend to zero. Thus $D_2(x, y) \le CT$. Infimizing over P gives the result. □

Lemma 4.3 (An initial comparison)**.** *There is a deterministic $C < \infty$ such that, almost surely,*

$$E(x, y) \le CD^U(x, y), \qquad x, y \in U. \tag{4.8}$$

Proof. Let

$$a_{z,r} = D(\partial B_{r/2}(z), \partial B_r(z)), \qquad b_{z,r} = \sup_{u,v\in\partial B_r(z)} D(u, v; A(z, r)). \tag{4.9}$$

The normalized pair $(a_{z,r}/s_r(z), b_{z,r}/s_r(z))$ has a law independent of z, r. Its first component is positive and its second is finite. Choose deterministic $a > 0$ and $b < \infty$ so that $a_{z,r} \ge as_r(z)$ and $b_{z,r} \le bs_r(z)$ except on an event of arbitrarily small probability. By Corollary 3.5, with probability tending to one uniformly over centers in each compact subset of U,

$$\sup_{u,v\in\partial B_r(z)} E(u, v; A(z, r)) \le b_{z,r} + as_r(z). \tag{4.10}$$

Choose the three error probabilities so that their sum is less than $1-p_*$. The comparison event (4.7) then holds with $C=(b+a)/a$ for all sufficiently small radii.

Both metrics inside the shell are determined by $H|_{A(z,r)}$. A constant shift of H multiplies both by the same factor. The crossing distance equals its internal-shell version by the subpath observation following (4.2). Thus the event is measurable with respect to (4.4). Lemma 4.2, with $D_1=D^U$ and $D_2=E$, proves (4.8). □

4.2 Annuli with a short connection on the inner sphere

Fix once and for all $\alpha\in(1/2,1)$, and define

$$\ell_{z,r}=D(\partial B_{\alpha r}(z),\partial B_r(z);A(z,r)),\qquad m_{z,r}=\sup_{u,v\in\partial B_{\alpha r}(z)}D(u,v;A(z,r)). \tag{4.11}$$

The ratio $m_{z,r}/\ell_{z,r}$ is finite, and its law does not depend on z,r.

Lemma 4.4 (Good annuli)**.** *There is a deterministic $M<\infty$, chosen independently of $\delta>0$, such that the following holds. For every $\delta>0$ and every compact set of centers in U, for all sufficiently small r the event $G_{z,r}$ defined by*

$$m_{z,r}\le M\ell_{z,r}, \tag{4.12}$$

$$E(u,v;A(z,r))\le D(u,v;A(z,r))+\delta\ell_{z,r},\quad u\in\partial B_{\alpha r}(z),\ v\in\partial B_r(z) \tag{4.13}$$

has probability at least p_*, uniformly over the centers. It is measurable with respect to (4.4). Consequently, the half-balls of these good annuli give the almost-sure coverings of Lemma 4.1.

Proof. Choose M so that (4.12) fails with probability less than $(1-p_*)/3$. This choice uses only the fixed law of $m_{z,r}/\ell_{z,r}$, not δ. Also choose $c>0$ so that $\ell_{z,r}\ge cs_r(z)$ except with probability less than $(1-p_*)/3$. Corollary 3.5, applied with error δc, makes the remaining failure probability less than $(1-p_*)/3$ at all sufficiently small scales. This gives the probability bound.

Locality shows that the event is determined by the field in the open shell. Both inequalities are unchanged by adding a constant to the field, so the event depends only on the shell restriction modulo constants, and therefore on (4.4). Lemma 4.1 applies. □

4.3 Improvement of the comparison constant

We do not assume that the whole-space metric is proper or geodesically complete. The following local fact suffices.

Lemma 4.5 (Local geodesics)**.** *Let D be a continuous length metric on $\mathbb{R}^d$, and let $K\Subset U$. There is a compact $K'\Subset U$ and a number $\rho>0$, possibly depending on D, such that each pair $x,y\in K$ with $|x-y|<\rho$ is joined by a D-geodesic contained in K'. For such pairs $D(x,y)=D(x,y;U)$.*

Proof. Choose a bounded open set O with $K\subset O\subset\overline{O}\Subset U$ and set $K'=\overline{O}$. Compactness gives $a=D(K,\partial O)>0$. Uniform continuity of D on $K\times K$ gives $\rho>0$ such that $D(x,y)<a/2$ for the indicated pairs. Almost-minimizing paths of length less than a cannot leave O. Parameterize such paths at constant speed on a common interval. They are equicontinuous for D and take values in the compact metric space $\overline{O}$. Arzelà–Ascoli and lower semicontinuity of length produce a minimizing path in $\overline{O}$. The same paths show equality with the internal distance in U. □

Proposition 4.6 (One-sided conformal comparison)**.** *Almost surely, $E\le D^U$ on $U\times U$.*

Proof. Let

$$C_* := \inf\{C > 0 : \mathbb{P}[E \le CD^U \text{ on } U \times U] = 1\}. \tag{4.14}$$

Lemma 4.3 makes this number finite. A decreasing sequence of admissible constants shows that the inequality also holds at C_*. Suppose that $C_* > 1$. Choose M from Lemma 4.4 and then set

$$\delta = \frac{C_* - 1}{2}, \qquad C' = \frac{MC_* + 1 + \delta}{M + 1} = C_* - \frac{C_* - 1}{2(M + 1)} < C_*. \tag{4.15}$$

All three constants are deterministic.

Work on the probability-one event on which $E \le C_*D^U$ and the good-annulus coverings hold along dyadic scales for a compact exhaustion. Fix $K \Subset U$ and use Lemma 4.5 to choose K' and ρ. Let $P : [0, T] \to K'$ be a D-geodesic between $x, y \in K$ with $|x - y| < \rho$, parameterized by D-length. Thus $T = D^U(x, y)$.

At a good covering scale ε, put $t_0 = 0$. Given $t_{j-1} < T$, choose a good ball $B_{r_j/2}(z_j)$ containing $P(t_{j-1})$, and let t_j be the first subsequent exit from $B_{r_j}(z_j)$, or T if the path has no such exit. As in Lemma 4.2, the positive crossing distances and the finite collection of balls imply termination in finitely many steps.

Consider a step whose outer ball contains neither x nor y. The path exits this ball at t_j, possibly with $t_j = T$. Let s_j be the last time before t_j at which P visits the inner sphere $\partial B_{\alpha r_j}(z_j)$. Since x is outside the outer ball and $P(t_{j-1})$ is inside the half-ball, there is an earlier time $a_j < t_{j-1}$ with $P(a_j)$ on the inner sphere. Geodesicity and (4.12) give

$$s_j - t_{j-1} \le s_j - a_j = D(P(a_j), P(s_j)) \le M\ell_{z_j,r_j} \le M(t_j - s_j). \tag{4.16}$$

The segment $P|_{[s_j,t_j]}$ lies in the closed radial region between the two spheres, hence in the open shell. Its internal D-distance equals its length $t_j - s_j$. By (4.13),

$$E(P(s_j), P(t_j)) \le (1 + \delta)(t_j - s_j). \tag{4.17}$$

For the preceding segment use $E \le C_*D^U$. The geodesic lies in U, so its internal distances also equal its subarc lengths. Writing $a = s_j - t_{j-1}$ and $b = t_j - s_j$, equations (4.16)–(4.17) imply

$$E(P(t_{j-1}), P(t_j)) \le C_*a + (1 + \delta)b \le C'(a + b), \tag{4.18}$$

since $a \le Mb$ and $1 + \delta < C_*$.

It remains to estimate steps whose balls contain an endpoint. Set

$$\omega_\varepsilon = \sup\{D(u, v) : u, v \in K',\ |u - v| \le 2\varepsilon\}. \tag{4.19}$$

Then $\omega_\varepsilon \to 0$. Each step has duration at most ω_ε, since its two endpoints lie in the same outer ball and P is a geodesic. If x lies in its outer ball, then $t_{j-1} = D(x, P(t_{j-1})) \le \omega_\varepsilon$. If y lies in its outer ball, then $T - t_{j-1} \le \omega_\varepsilon$. Every step starting in $(\omega_\varepsilon, T - \omega_\varepsilon)$ therefore satisfies (4.18). These steps form a consecutive block. The initial and final omitted intervals have total duration at most $4\omega_\varepsilon$; if the block is empty, $T \le 4\omega_\varepsilon$.

Using $E \le C_*D^U$ on the omitted intervals and summing (4.18), we obtain

$$E(x, y) \le C'D^U(x, y) + 4C_*\omega_\varepsilon. \tag{4.20}$$

Letting the dyadic scale tend to zero gives $E(x, y) \le C'D^U(x, y)$ simultaneously for the indicated near pairs in K. The construction on the fixed event was deterministic and uniform in these pairs.

Every D-rectifiable curve in U is contained in some compact set of the exhaustion. Divide such a curve into finitely many pieces whose endpoints satisfy the local comparison. The triangle inequality gives an E-distance bounded by C' times its D-length. Infimizing over curves proves $E \le C'D^U$ everywhere on the same probability-one event. This contradicts (4.14), since $C' < C_*$. Hence $C_* \le 1$. □

5 Conformal covariance on domains

We complete the passage from the one-sided comparison for a whole-space field to the domain-family identity. Two deterministic observations make the passage precise.

First, if $d_1 \le d_2$ are continuous length metrics on the same domain, then

$$e^{\xi f} \cdot d_1 \le e^{\xi f} \cdot d_2 \qquad (f \in C(U)). \tag{5.1}$$

For a d_2-rectifiable path, the length inequality on every subinterval shows that its d_1-length measure is dominated by its d_2-length measure. Integrating the positive continuous weight gives the weighted length inequality. Infimizing over such paths proves (5.1). Second, Weyl multiplication commutes with pullback of a metric under a homeomorphism; the weight is composed with that homeomorphism. These facts follow from path lengths and do not presume conformal covariance.

Proof of Theorem 1.3. Proposition 2.4 constructs the unique domain family satisfying Axioms I–III. By Lemma 3.1, the transformed field h^ϕ is in the same local field class, so both sides of (1.15) are defined.

Proposition 4.6 gives the one-sided inequality for $h = H|_U$. Equations (5.1) and (1.9) extend it to $h = H|_U + f$ for any continuous f, including one dependent on H. Indeed, the added term in the transformed field is $f \circ \phi^{-1}$, whose pullback is precisely f.

For a general locally LGF-type h, take a countable cover of U by relatively compact balls with local representations of the form (1.4). Apply the preceding one-sided comparison to the restriction of ϕ on each ball. By locality, it compares the internal metrics of D_h^ϕ and D_h there. Their path lengths consequently satisfy the same inequality on every path contained in one such ball. Subdividing any rectifiable path into finitely many pieces in the cover, and then infimizing, gives

$$D_h^\phi \le D_h \quad \text{on } U \times U \tag{5.2}$$

almost surely. Each representation is only used to establish an almost-sure assertion about the Borel functions of h on that ball, so different auxiliary couplings cause no ambiguity.

Apply (5.2) to h^ϕ on V and the inverse map. The Jacobian chain rule gives the distributional identity

$$(h^\phi)^{\phi^{-1}} = h. \tag{5.3}$$

There is no additive constant in this identity, since the transformed fields were not independently normalized. The inverse-map inequality is exactly $D_h \le D_h^\phi$. Together with (5.2), it proves (1.15).

Only countable covers, compact exhaustions, and sequences of scales were intersected. Thus the equality holds simultaneously for all points, for each fixed conformal map and each fixed admissible input law, as required. Uniqueness was already proved in Proposition 2.4. □

The role of the zero-boundary field is now explicit. It supplies the domain field used in the definition and is compatible with the whole-space field through a smooth complementary distribution. The metric on that field is constructed by (1.14), not postulated. Conformal covariance is the additional conclusion proved after this construction. The argument leaves the existence of the strong whole-space input, including its consistent locality versions, to a separate theorem.

References

[1] A. A. Contreras Hip and Z. Zhuang. *Weak exponential metrics for high-dimensional log-correlated Gaussian fields*. Preprint, 2025. arXiv:2512.06292.

[2] S. Dipierro and H.-C. Grunau. *Boggio's formula for fractional polyharmonic Dirichlet problems*. Ann. Mat. Pura Appl. **196** (2017), 1327–1344. arXiv:1606.03110.

[3] B. Duplantier, R. Rhodes, S. Sheffield, and V. Vargas. *Log-correlated Gaussian fields: an overview*. Preprint, 2014. arXiv:1407.5605.

[4] E. Gwynne and J. Miller. *Conformal covariance of the Liouville quantum gravity metric for* $\gamma \in (0, 2)$. Ann. Inst. H. Poincaré Probab. Statist. **57** (2021), 1016–1031. arXiv:1905.00384.

[5] A. Lodhia, S. Sheffield, X. Sun, and S. S. Watson. *Fractional Gaussian fields: a survey*. Probab. Surv. **13** (2016), 1–56. arXiv:1407.5598.